# Shooting Stars From Halley's Comet Light Up the Autumn Sky

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Halley's Comet is more than 3 billion miles (4.8 billion kilometers) from Earth right now, but pieces it left behind are still putting on a show. On October 21, **astrophotographer** Tim McCollum captured three bright Orionid meteors streaking across the sky above Toccoa, Georgia.

The Orionid meteor shower happens each autumn, when Earth passes through the trail of dust and **debris** shed by Halley's Comet. As the tiny particles burn up in our atmosphere, they become shooting stars. In McCollum's picture, the meteors appear to race away from a point near Betelgeuse, the red giant star in the **constellation** Orion. That point is called the **radiant**.

Catching a shooting star on camera takes skill, luck and patience. McCollum photographed the sky between 3 and 4 a.m. using a Canon camera on a star tracker, taking many 30-second exposures. He then combined them into one **composite** image with astrophoto software.

The finished picture holds more than meteors. The familiar stars of Orion's Belt shine near the glow of the Orion Nebula, one of the closest major star-forming regions to Earth. The Rosette Nebula also appears, tucked into the dense star fields and dust lanes of the Milky Way, which cuts diagonally across the image.

The Orionid shower stays active until November 7, so skywatchers still have chances to spot one of Halley's shooting stars on the next few clear nights. Earth is also passing through **debris** from Comet 2P/Encke, so a lucky viewer might even catch a bright Taurid meteor, sometimes nicknamed a Halloween Fireball.

## Word Treasure

**astrophotographer** -- A person who takes photographs of stars, planets and other objects in space.

**debris** -- Small broken pieces left behind by something.

**radiant** -- The point in the sky that meteors in a shower seem to race away from.

**composite** -- Made by combining several separate pictures or parts into one.

**constellation** -- A pattern of stars in the sky with a name, such as Orion.

## Background you need

A comet sheds dust and small pieces as it travels, leaving a trail of debris behind it in space.

When Earth passes through that trail, the tiny pieces burn up in the atmosphere as meteors, which people call shooting stars.

The Orionid meteors seem to come from near the constellation Orion, which is why the shower is named for Orion.

## Break it down

**LEAD:** Debris from Halley's Comet is putting on a shooting-star show even though the comet is billions of miles away

**KEY EVENT:** Tim McCollum captured three bright Orionid meteors above Toccoa, Georgia, on October 21

+- WHO: Astrophotographer Tim McCollum

+- HOW: Many 30-second exposures on a star tracker were combined into one composite image

**EVIDENCE:** The picture shows Orion's Belt, the Orion Nebula, the Rosette Nebula and the Milky Way

**OPEN QUESTION:** Whether viewers will also catch a bright Taurid meteor, nicknamed a Halloween Fireball

## Why it matters

Meteor showers let anyone watching the sky see Earth's journey through a comet's debris trail with their own eyes. This shower stays active until November 7, so families still have clear nights to look for one.

## Test what you remember

**1. How far is Halley's Comet from Earth right now?**

- (A) More than 3 billion miles (4.8 billion kilometers)
- (B) Exactly at the Moon
- (C) Less than one million miles
- (D) About 3 million miles

**2. On what date did Tim McCollum capture the three Orionid meteors?**

- (A) October 21
- (B) October 1
- (C) November 7
- (D) October 31

**3. Why does the Orionid meteor shower happen each autumn?**

- (A) Orion's stars fall to Earth
- (B) The Moon blocks the comet
- (C) A spacecraft drops dust into the sky
- (D) Earth passes through the trail of dust and debris shed by Halley's Comet

**4. What is the radiant?**

- (A) A type of camera
- (B) A bright Taurid meteor
- (C) The glow of the Orion Nebula
- (D) The point near Betelgeuse that the meteors appear to race away from

**5. How did McCollum make his finished picture?**

- (A) He photographed from a spacecraft
- (B) He used a single one-second photo
- (C) He combined many 30-second exposures into one composite image
- (D) He painted the meteors afterward

**6. Until when does the Orionid shower stay active?**

- (A) Halloween night only
- (B) November 7
- (C) December 25
- (D) October 23

Answer key (try the quiz first!): 1: A 2: A 3: D 4: D 5: C 6: B

# Questions to think about

## MY THOUGHTS (at least 20 words):